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Bank Smart Security System

Embedded Systems Project

Team Members

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Course: Embedded Systems

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 **GitHub Repository**

1 Introduction

Banks and restricted facilities require reliable and intelligent security systems capable of detecting unauthorized access, monitoring abnormal conditions, and sending immediate alerts. Traditional security systems usually depend on human supervision, passwords, keys, or simple alarm circuits. These methods may fail to provide fast automated response, remote monitoring, or evidence collection during emergencies.

This project presents a smart bank security system integrating embedded systems, machine learning, computer vision, and IoT technologies. The proposed system combines a PIC16F877A microcontroller, Raspberry Pi, camera module, PIR sensor, voltage sensor, relay-controlled solenoid lock, buzzer, LED indicators, cloud services, and a mobile application. The system provides real-time monitoring, automatic safety actions, and remote notifications.

The implemented system can:

- Detect authorized and unauthorized persons using face recognition.
- Monitor motion in restricted areas using a PIR sensor.
- Monitor power conditions using a voltage sensor.
- Automatically control a solenoid door lock through a relay module.
- Trigger local alarms using a buzzer and LED indicators.
- Send real-time alerts and captured images to a mobile application.

2 Literature Review

Several recent works have proposed IoT-based security systems using cameras, sensors, cloud communication, and machine-learning methods. These systems show that combining embedded hardware with computer vision can improve access control and intrusion detection compared to traditional lock-and-key or password-based security.

Margapuri et al. proposed *PiBase*, an IoT-based security system using Raspberry Pi, PIR motion sensor, Pi camera, Android application, and Google Firebase. Their system detects human intrusion using PIR and camera input, applies Haar cascade for face detection and LBPH for face recognition, and sends mobile notifications using Firebase Cloud Messaging. This work is closely related to our project because it combines Raspberry Pi, PIR sensing, camera-based recognition, cloud database services, and mobile alerts [1].

Nguyen et al. proposed a smart security system based on face recognition for authenticated access. Their system architecture includes a camera node, cloud server, smartphone, and smart devices. The Raspberry Pi camera node detects faces and performs image processing using Dlib and TensorFlow, while the cloud server stores data and supports model training. This work supports the use of Raspberry Pi and machine learning for automated access-control systems [2].

Kumar et al. discussed smart face recognition using IoT and machine learning. Their work describes the general pipeline of face recognition systems, including data acquisition, face detection, face alignment, feature extraction, face recognition, and decision making. The paper also highlights the importance of cameras and sensors for real-time smart security applications [3].

Krishnaveni and Karthick proposed an IoT-based bank locker security system with face recognition. Their system uses Raspberry Pi camera-based face recognition as part of a multi-stage authentication process for bank locker access. This paper is directly related to our application domain because it targets bank security and restricted locker access using face recognition and IoT communication [4].

Compared with the reviewed systems, our project combines Raspberry Pi-based face recognition with a PIC16F877A microcontroller for low-level hardware control. The PIC handles sensor readings and actuator control, while the Raspberry Pi handles camera processing, face recognition, and cloud communication. This separation improves modularity and makes the system suitable for both embedded hardware implementation and intelligent monitoring.

3 Used Components

Component	Function
PIC16F877A	Main microcontroller for sensor interfacing and actuator control.
Raspberry Pi	Runs camera processing, face recognition, cloud communication, and system logic.
Camera Module	Captures face images and evidence images during intrusion events.
PIR Sensor	Detects motion in restricted areas.
Voltage Sensor Module	Measures power supply condition and sends analog output to PIC RA0.
2-Channel Relay Module	Switches high-power loads such as solenoid lock, alarm, or lighting.
Solenoid Lock	Provides automatic door locking/unlocking.
Logic Level Converter	Converts UART signals between PIC 5V logic and Raspberry Pi 3.3V logic.
Buzzer	Provides audible alarm during unauthorized access or power fault.
LED Indicators	Show system status such as alert, transmission, or lock state.
Mobile Application	Displays alerts, images, and system status to the manager/user.
Supabase/Firebase	Provides cloud database, image storage, and notification support.

Table 1: List of used components and their functions.

4 Input/Output Diagram

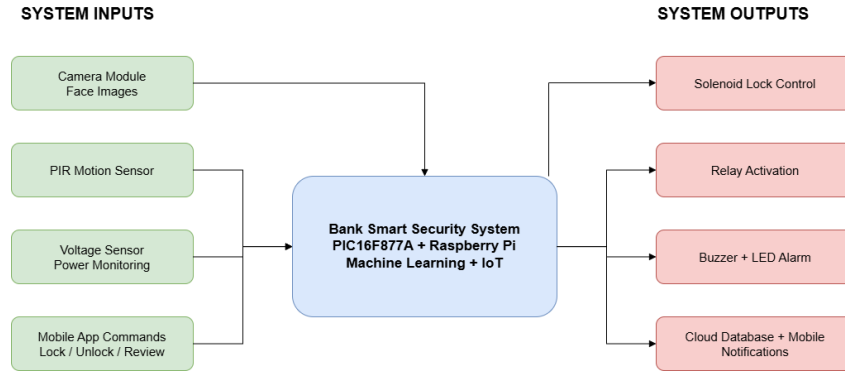


Figure 1: Input/output diagram of the proposed system.

5 System Architecture Schematic

The system architecture consists of two main processing layers. The first layer is the embedded control layer based on the PIC16F877A microcontroller. This layer reads the voltage sensor and PIR sensor and controls output actuators such as the relay, buzzer, and LED. The second layer is the intelligent processing layer based on the Raspberry Pi. This layer runs face recognition, communicates with the cloud database, and sends notifications to the mobile application.

The PIC and Raspberry Pi communicate through UART. Since the PIC operates at 5V logic while the Raspberry Pi operates at 3.3V logic, a logic level converter is used to protect the Raspberry Pi GPIO pins.

Sensors → PIC16F877A ↔ Raspberry Pi → ML Processing → Cloud Database → Mobile App

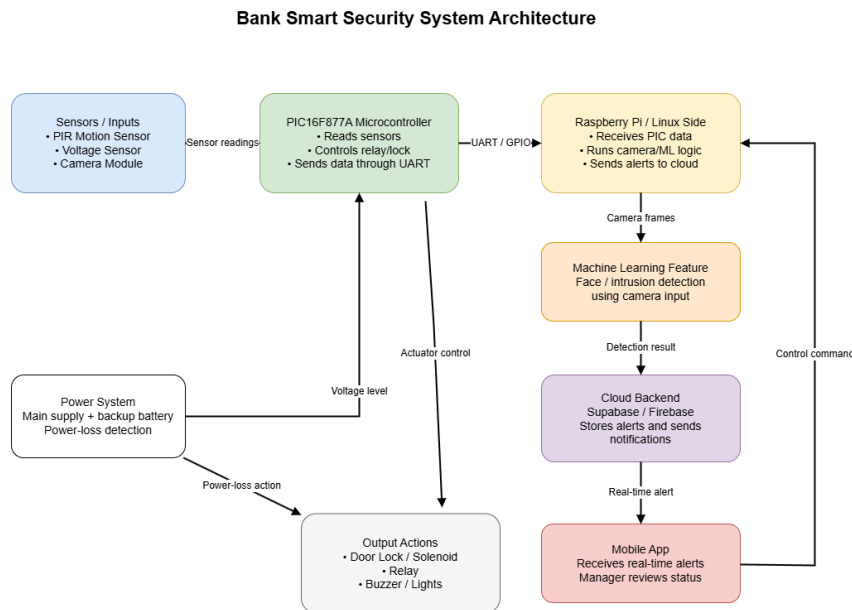


Figure 2: System architecture schematic.

6 Electrical Schematic and PCB Design

The electrical schematic was designed using KiCad. The design includes the PIC16F877A, Raspberry Pi header, 12-pin logic level converter, 2-channel relay module, solenoid lock connector, PIR sensor connector, voltage sensor connectors, buzzer, LED indicators, crystal oscillator, and power connectors.

The main electrical connections are:

- Voltage sensor output connected to PIC RA0.
- PIR sensor output connected to PIC RB1.
- Relay input connected to PIC RD0.
- Relay input 2 connected to PIC RD2.
- Buzzer/LED status connected to PIC RD3.
- PIC UART TX/RX connected to Raspberry Pi UART through the logic level converter.
- Relay output terminals connected to the solenoid lock power path.
- Common ground is shared between PIC, Raspberry Pi, sensors, relay module, and power supplies.
- Logic convertor was used for uart communication and upconverting or downconverting 3.3 V and 5 V.

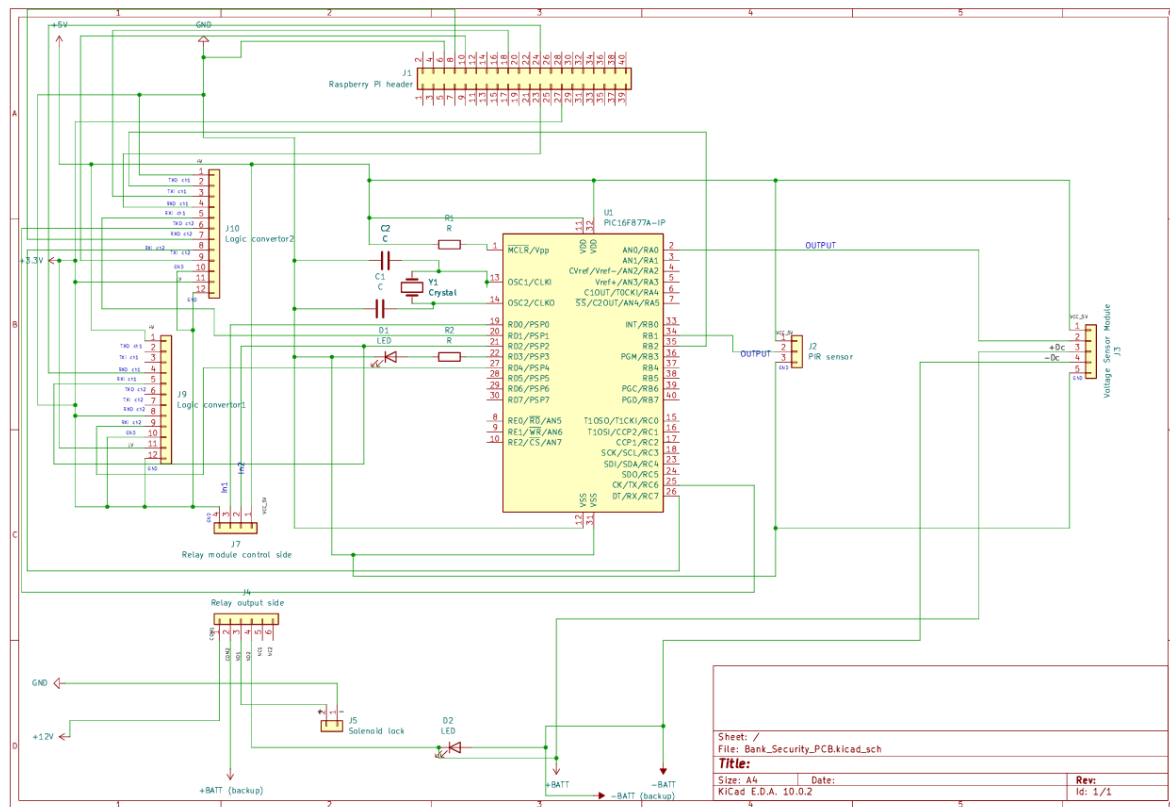


Figure 3: Electrical schematic designed in KiCad.

7 Implemented Safety-Critical Features

7.1 Feature 1: Face Recognition Access Control

The Raspberry Pi camera captures a face image and runs the face recognition pipeline. If the detected person is authorized, the system allows access. If the person is unknown or unauthorized, the door remains locked and an alert is sent to the mobile application.

Pseudocode

```
Start camera
Capture image frame
Detect face in the image
Generate face embedding
Compare embedding with registered database

IF face is recognized as authorized THEN
    Send unlock command
    Deactivate alarm
    Log authorized access
ELSE
    Keep solenoid lock closed
    Activate buzzer and LED alert
    Capture evidence image
    Upload alert to cloud database
    Send mobile notification
END IF
```

7.2 Feature 2: Intrusion Detection Using PIR Sensor

The PIR sensor monitors motion inside restricted areas. If motion is detected after working hours or while the area is locked, the system treats the event as a possible intrusion.

Pseudocode

```
Read PIR sensor output

IF PIR output is HIGH THEN
    Send motion event to Raspberry Pi
    Turn on LED indicator
    Activate buzzer
    Capture image using camera
    Upload image and alert details to cloud
    Send intrusion notification to mobile app
END IF
```

7.3 Feature 3: Voltage Monitoring and Power Protection

The voltage sensor measures the supply voltage. If the measured voltage drops below a defined threshold, the system generates a power warning and triggers the backup response.

Pseudocode

```
Read analog voltage from RA0
Convert ADC value to voltage level

IF voltage is below threshold THEN
    Activate warning LED
    Activate buzzer
    Send power fault message to Raspberry Pi
    Upload power warning to cloud database
    Notify mobile application
ELSE
    Continue normal operation
END IF
```

7.4 Feature 4: User Door Locking and Unlocking

This feature allows the user or manager to control the door lock remotely from the mobile application. The command is sent to the Raspberry Pi, then forwarded to the PIC microcontroller, which controls the relay connected to the solenoid lock.

Pseudocode

```
Wait for command from mobile application

IF command == "LOCK" THEN
    Raspberry Pi sends LOCK command to PIC through UART
    PIC activates relay state for locked mode
    Solenoid lock keeps the door locked
    Update door status in cloud database
    Display "Door Locked" on mobile app

ELSE IF command == "UNLOCK" THEN
    Raspberry Pi sends UNLOCK command to PIC through UART
    PIC changes relay state for unlocked mode
    Solenoid lock releases the door
    Update door status in cloud database
    Display "Door Unlocked" on mobile app

END IF
```

8 Implementation Phase

The implementation phase included hardware assembly, embedded programming, Raspberry Pi software development, face recognition model integration, cloud database setup, and mobile application implementation.

8.1 Hardware Implementation

- PIC16F877A was used for low-level control.
- Sensors were connected to PIC input pins.
- Relay module was connected to PIC output pins.
- Raspberry Pi was connected to the camera and cloud services.
- Logic level converter was used for safe UART communication.

8.2 Software Implementation

- PIC firmware reads sensor values and controls actuators.
- Raspberry Pi code processes camera input and runs face recognition.
- Cloud database stores alerts, images, timestamps, and device status.
- Mobile application displays real-time alerts and system updates.

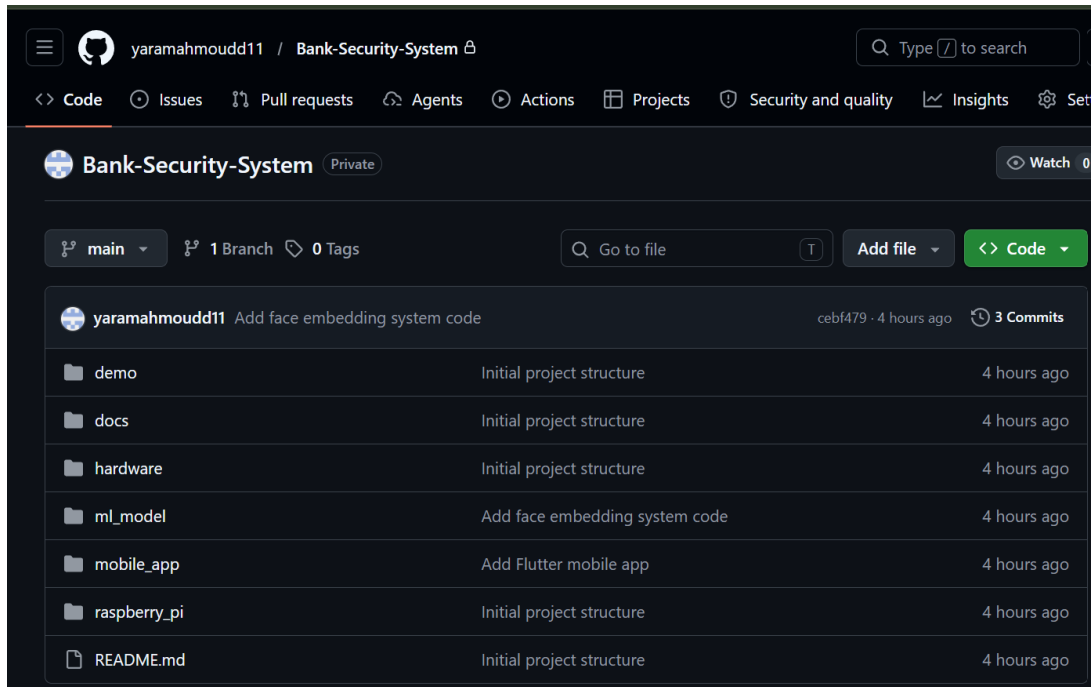


Figure 4: Documented and replicable codebase.

9 Integration Phase

The integration phase connected all modules into one complete system. Sensor readings from the PIC were sent to the Raspberry Pi through UART. The Raspberry Pi processed camera images, checked the recognition result, updated the cloud database, and triggered notifications to the mobile application. The output actions such as door locking, buzzer activation, and LED status were controlled through the relay and PIC output pins.

Testing was performed to verify:

- Correct PIR motion detection.
- Correct voltage sensor reading.
- Correct UART communication between PIC and Raspberry Pi.
- Correct face recognition response.
- Correct relay and solenoid lock operation.
- Correct mobile notification delivery.



Figure 5: PIC to Raspberry Pi communication test.

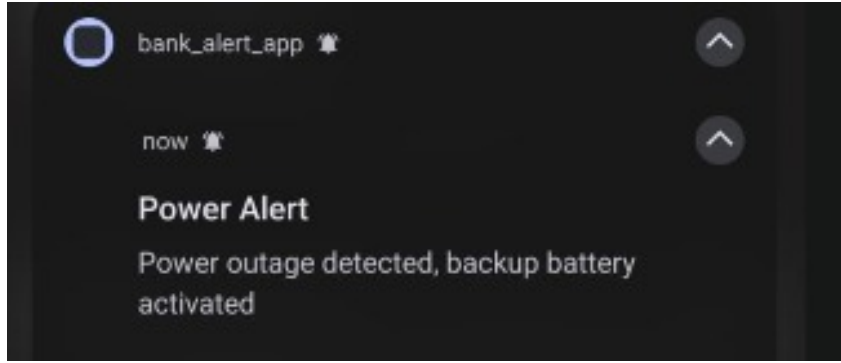


Figure 6: Mobile application alert display.

10 Demo Phase

The demo phase validates the full system operation under realistic scenarios.

10.1 Demo Scenario 1: Authorized Access

1. Authorized person stands in front of the camera.
2. The Raspberry Pi recognizes the face.
3. The system allows access and keeps alarm outputs inactive.

10.2 Demo Scenario 2: Unauthorized Person

1. Unknown person appears in front of the camera.
2. The system classifies the face as unauthorized.
3. The solenoid lock remains locked.
4. Mobile application receives an alert with image evidence.

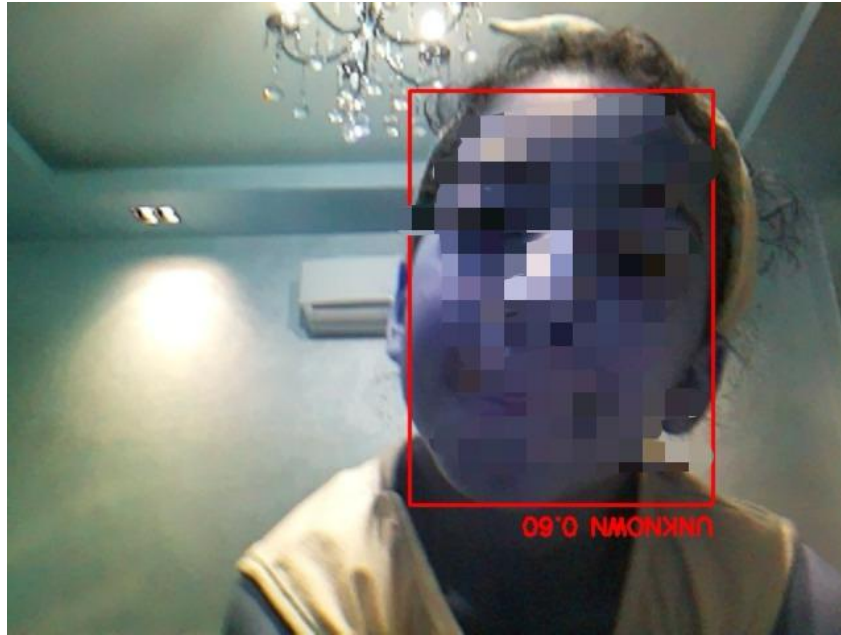


Figure 7: Unauthorized access demo.

10.3 Demo Scenario 3: Motion Detection

1. PIR sensor detects motion in the restricted area.
2. Raspberry Pi captures an image.
3. The system turns on the lights.
4. Alert is uploaded to the cloud and shown in the mobile application.
5. the camera takes a pic of the intrusion location and sends it to the end user.

10.4 Demo Scenario 4: Power Fault Detection

1. Voltage drops below the threshold.
2. PIC detects the fault using ADC reading.
3. Warning output is activated.
4. Mobile application receives a power alert.

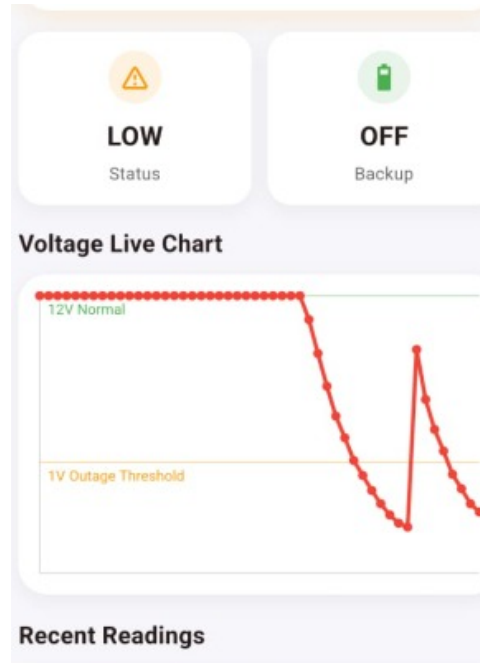


Figure 8: Power monitoring demo.

11 Conclusion

This project demonstrates an intelligent bank security system that combines embedded systems, IoT, cloud communication, and machine learning. The system detects unauthorized access, monitors motion and power conditions, controls a solenoid lock, and sends real-time alerts to a mobile application.

The design improves safety by combining automatic hardware response with remote monitoring. The use of a PIC microcontroller for hardware control and Raspberry Pi for machine-learning processing makes the system modular, practical, and suitable for safety-critical applications.

References

- [1] V. Margapuri, N. Penumajji, and M. Neilsen, “PiBase: An IoT-based Security System Using Google Firebase and Raspberry Pi,” 2021.
- [2] T. Nguyen, B. Lakshmanan, and W. Sheng, “A Smart Security System with Face Recognition,” arXiv preprint arXiv:1812.09127, 2018.
- [3] A. Kumar, S. Kalumbi, M. K. Gupta, and P. Rao, “Smart Face Recognition Using IoT and Machine Learning,” *International Journal of Research Publication and Reviews*, vol. 4, no. 5, pp. 4161–4171, 2023.
- [4] K. A. Krishnaveni and K. Myilavahanan, “IoT based Bank Locker Security System with Face Recognition,” *Grenze International Journal of Engineering and Technology*, 2023.